

JANE ODUM

📍 ATHENS, UNITED STATES 📞 7623168444

◦ DETAILS ◦

Athens
United States
7623168444
janeodum41@gmail.com

◦ LINKS ◦

LinkedIn
Website

◦ SKILLS ◦

Machine Learning & Deep Learning

Scalable System Design &
Distributed Computing

Natural Language Processing &
Computer Vision

C/C++

Python

Java

Tensor flow

pyTorch

LLM

◦ HOBBIES ◦

Painting, Dancing, Working out
and travelling.



EMPLOYMENT HISTORY

Machine Learning Engineer (Intern) at Stripe, South San Francisco

May 2025 — Aug 2025

- Developed and deployed ML models for secret key leak detection, cutting detection latency from 4.5 days to under 1 hour and preventing multi-million-dollar annual fraud losses.
- Engineered large-scale feature pipelines in Spark/Databricks (500M+ records across Iceberg tables), integrated with Shepherd and Airflow for production-ready workflows.
- Expanded detection coverage by onboarding new API-call signals (OAuth, Connect), boosting recall on confirmed leaks to 94% while ensuring system observability with custom dashboards.

Graduate Teaching Assistant at University of Georgia, Athens

Jan 2021 — Present

- Collaborated with Prof. Shelby Funk and Dr. Michael Cotterell in delivering course content for Theory of Computing and Human-Computer Interaction.
- Evaluated and graded assignments and exams with adherence to university grading standards, providing timely and constructive feedback.

Software Developer(Intern) at Orbital Schools, Ilorin

May 2018 — Aug 2018

- Spearheaded the development of dynamic websites utilizing the LAMP stack, enhancing client digital presence with Linux, PHP 5, CSS, and MySQL.
- Executed seamless integration of payment processing capabilities by embedding PayStack and other APIs into Orbital Schools' web platform.



EDUCATION

PhD Computer Science, University of Georgia, Athens

May 2026

- **Research Interest:** Time Series Forecasting, Generative AI, NLP, Content understanding using large Language models and multi-modal machine learning.
- **Coursework:** Machine Learning, Algorithms, Software engineering, Data mining, Trustworthy Machine Learning, Databases, Advanced Robotics, Human-Computer Interaction and Advanced Representation Learning.

Software engineering, 42 Silicon Valley, Fremont

Dec 2019 — Dec 2020

Bsc Computer Science, University of Ilorin, Ilorin

Oct 2016 — Oct 2019

★ PUBLICATION

- **Adaptive Quantile Guidance in Diffusion Models: Multi-Dataset Learning for Pandemic Time Series.** ICMLA 2025 (Accepted).
 - **Developed AQDiff**, a diffusion-based framework for epidemiological forecasting, leveraging multi-dataset pre-training (CDC influenza, COVID-19, RSV) and adaptive quantile guidance to dynamically adjust prediction intervals. Achieved **73.3% lower MAE** than state-of-the-art models (CSDI, PatchTST) across pandemic scenarios.
 - **Introduced real-time error feedback** via exponentially weighted residual statistics, enhancing volatility adaptation with <10% computational overhead. Demonstrated robustness in sparse data methods and statistically significant improvements during outbreak surges.

★ PROJECTS

- **Prompt Fusion in Stable Diffusion**
 - Collaborated on the Prompt Fusion project to optimize image generation by consolidating positive and negative prompts into a single efficient pass.
 - Utilized LPIPS and Siamese networks for image evaluation, and BERT for prompt similarity, significantly improving optimization time and image-prompt alignment.
- **HumorAI: Enhancing LLM for Implicit Humor and Sentence Analysis**
 - Engineered an advanced LLM model for sentiment analysis, integrating humor recognition to refine sentence similarity scores and enhancing the model's language understanding capabilities.
 - Utilized contrastive loss to train the SBERT and BERT models, distinguishing between similar and dissimilar sentence pairs for sentiment analysis.
 - Applied this enhanced model to a chatbot and Twitter sentence similarity tasks, achieving a noticeable 15% improvement in performance accuracy compared to other existing models, demonstrating its effectiveness in practical applications.
- **StockCluster: Enhancing Investment Portfolios**
 - Devised a clustering algorithm for stock market analysis, and achieved a 6.89% ROI increase by optimizing portfolio diversification.
- **Risk-Aware Task Allocation in multi-robot system**
 - Developed a risk-aware multi-robot task allocation model, factoring in task dependencies and robot characteristics to optimize performance while minimizing risk, increasing operational efficiency by 40% and improving task completion rates
 - Expanded the approach to tackle the ST-MR-TA challenge, improving real-world task allocation efficiency.

✎ LEADERSHIP

- **GOOGLE DEVELOPER STUDENT LEAD at Google, Ilorin**
Jul 2018 — Aug 2019
 - As a Google Developer Student Lead, guided 300+ students across 11 African countries in web and mobile development, focusing on local challenges.
- **UNITED AMBASSADORS (MUN). NEW YORK at Model United Nations, New York City**
Aug 2018 — Aug 2018
 - Represented the US as a UN Ambassador, advocating for global peace and spearheading immigrant and refugee rights discussions among 55 international ambassadors.